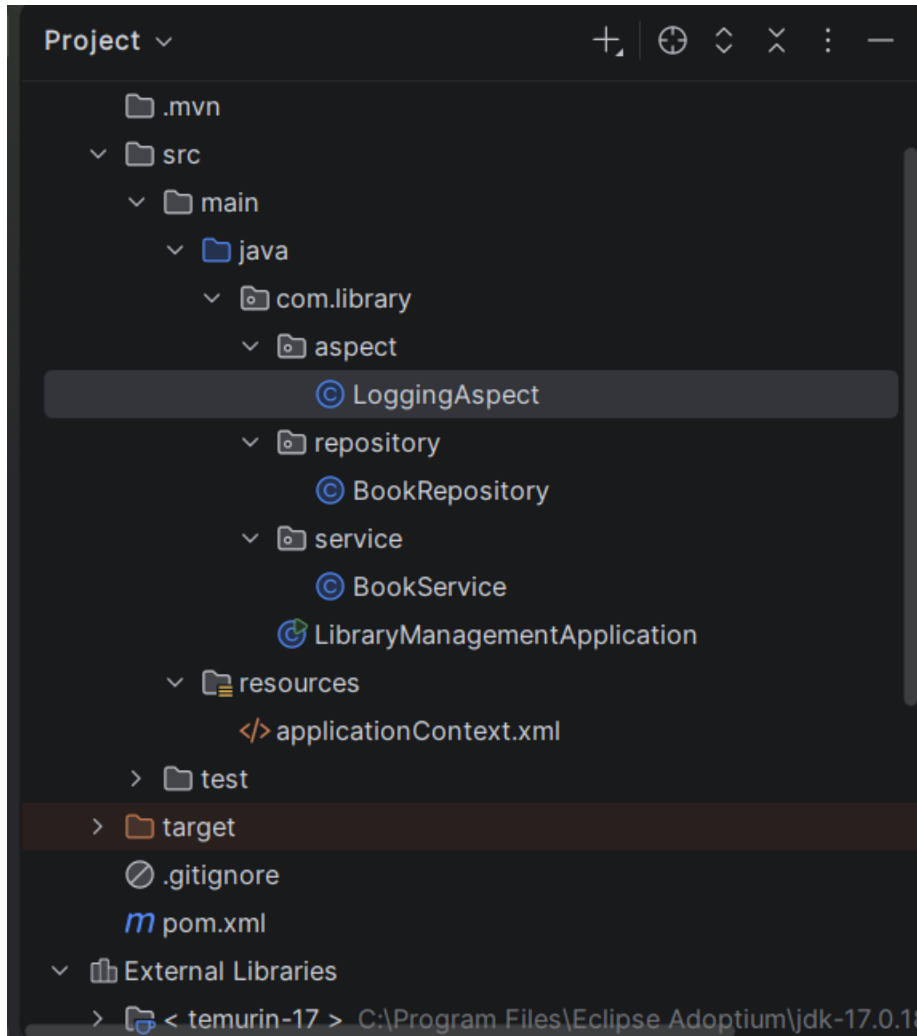


Exercise 3: Implementing Logging with Spring AOP

Scenario: The library management application requires logging capabilities to track method execution times.

Folder Structure:



Code:

LoggingAspect.java

```
package com.library.aspect;

import org.aspectj.lang.ProceedingJoinPoint;
import org.aspectj.lang.annotation.Around;
import org.aspectj.lang.annotation.Aspect;

@Aspect
public class LoggingAspect {

    @Around("execution(* com.library.service.*.*(..))")
```

```

public Object logExecutionTime(ProceedingJoinPoint joinPoint)
    throws Throwable {

    long startTime = System.currentTimeMillis();

    Object result = joinPoint.proceed();

    long endTime = System.currentTimeMillis();

    System.out.println(
        joinPoint.getSignature().getName()
            + " executed in "
            + (endTime - startTime)
            + " ms");

    return result;
}
}

```

BookRepository.java

```

package com.library.repository;

public class BookRepository {

    public void displayBook() {

        System.out.println("Fetching book details from repository.");

    }

}

```

BookService.java

```

package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void displayService() {

        System.out.println("Book Service is working.");

        bookRepository.displayBook();
    }

}

```

```
}  
}
```

LibraryManagementApplication.java

```
package com.library;  
  
import com.library.service.BookService;  
import org.springframework.context.ApplicationContext;  
import org.springframework.context.support.ClassPathXmlApplicationContext;  
  
public class LibraryManagementApplication {  
  
    public static void main(String[] args) {  
  
        ApplicationContext context =  
            new ClassPathXmlApplicationContext(  
                "applicationContext.xml");  
  
        BookService service =  
            context.getBean("bookService",  
                BookService.class);  
  
        service.displayService();  
  
    }  
}
```

Output:

The screenshot shows an IDE with the following components:

- Project Explorer:** Displays the project structure. The file `LibraryManagementApplication` is selected under the `com.library` package.
- Editor:** Shows the source code of `LibraryManagementApplication.java`. The code is identical to the one provided in the previous block.
- Run Console:** Shows the output of the application. The output is:
Book Service is working.
Fetching book details from repository.
displayService executed in 0 ms
Process finished with exit code 0

